

Second messenger c-di-AMP regulates multiple antibiotic sensitivity pathways in *Mycobacterium smegmatis* by discrete mechanisms

Aditya Kumar Pal¹, Dipankar Ghorai¹, Xueliang Ge², Biplab Sarkar¹, Amit Kumar Sahu³, Vikas Chaudhary¹, Ruchi Jhawar¹, Suparna Sanyal^{1,2}, Mahavir Singh¹, Anirban Ghosh^{1,*}

¹Molecular Biophysics Unit, Indian Institute of Science, Bangalore 560012, India

²Department of Cell and Molecular Biology, Uppsala University, Uppsala SE-75124, Sweden

³Department of Microbiology and Cell Biology, Indian Institute of Science, Bangalore 560012, India

*Corresponding author. Molecular Biophysics Unit, Indian Institute of Science, CV Raman road, Bangalore-560012, India. E-mail: ganirban@iisc.ac.in

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Abstract

One of the debilitating causes of high mortality in the case of tuberculosis and other bacterial infections is the resistance development against standard drugs. There are limited studies so far to describe how a bacterial second messenger molecule can directly participate in distinctive antibiotic tolerance characteristics of a cell in a mechanism-dependent manner. Here we show that intracellular cyclic di-AMP (c-di-AMP) concentration can modulate drug sensitivity of *Mycobacterium smegmatis* by interacting with an effector protein or interfering with the 5'-UTR regions in mRNA of the genes and thus causing transcriptional downregulation of important genes in the pathways. We studied four antibiotics with different mechanisms of action: rifampicin, ciprofloxacin, erythromycin, and tobramycin and subsequently found that the level of drug sensitivity of the bacteria is directly proportional to the c-di-AMP concentration inside the cell. Further, we unraveled the underlying molecular mechanisms to delineate the specific genes and pathways regulated by c-di-AMP and hence result in differential drug sensitivity in *M. smegmatis*.

Keywords: second messenger; mycobacteria; antibiotics; regulation; phenotypes; mechanism

Introduction

Antimicrobial resistance poses a huge threat to human civilization where bacterial infections cannot be cured using the existing antibiotics arsenal. Over the years, significant efforts have been made to reveal the specific cellular factors and pathways responsible for antibiotic resistance in bacteria (Munita and Arias 2016, Peterson and Kaur 2018, Reygaert 2018). Nucleotide second messenger (NSM) signaling is a vast area of research where small-molecule messengers (mostly nucleotide derived) have been shown to act as an intermediate player to link primary cellular or environmental cues to a regulatory effect, often associated with distinct phenotypes. Previously it has been found that second messenger (p)ppGpp could induce antibiotic resistance and persistence in many bacteria by diverse mechanisms (Rodionov and Ishiguro 1995, Pomares et al. 2008, Koskiniemi et al. 2011, Aedo and Tomasz 2016, Strugeon et al. 2016, Martins et al. 2018). The second messenger c-di-GMP has been studied for modulating the sensitivity of planktonic cells to different antibiotics, such as ethionamide in *Mycobacterium tuberculosis* (Zhang et al. 2017), imipenem in *Pseudomonas aeruginosa* (Nicastro et al. 2014), and erythromycin in *M. smegmatis* (Gupta et al. 2015). However, knowledge and understanding about how other second messenger molecules, including cyclic di-AMP (c-di-AMP) could impact the antibiotic sensitivity and resistance phenotypes of a cell in a target-dependent manner is still lacking. One recent study highlighted the novel c-di-AMP driven drug phenotypes in mycobacteria (Chaudhary et al. 2023) and another

study deciphered the mechanistic role of second messenger c-di-AMP determining drug resistance phenomenon in *M. smegmatis* (Pal and Ghosh 2022), which is often used as a surrogate to *M. tuberculosis* to evaluate drug leads (Shiloh 2010, Lelovic et al. 2020). A recent study highlighted c-di-AMP's possible role in determining heteroresistance phenotypes in *Staphylococcus aureus* (Heidarian et al. 2024). Though there were some previous studies of c-di-AMP regulating beta-lactam resistance in Gram-positive bacteria (Corrigan et al. 2011, Griffiths and O'Neill 2012, Luo and Helmann 2012, Witte et al. 2013, Kobras et al. 2023), the mechanism was not clear, and the effect seems to be indirect. Throughout this study, we used two previously characterized *M. smegmatis* strains with altered c-di-AMP levels: c-di-AMP null mutant (Δ disA) and a c-di-AMP hyperaccumulating mutant (Δ pde) along with the WT and respective complemented strains. Previously, it was reported how the overproduction of c-di-AMP makes cells sensitive to rifampicin and vancomycin (Chaudhary et al. 2023). In this study, we seek to understand the specific underlying mechanisms linking high c-di-AMP concentration to antibiotic sensitivity with rifampicin. In addition to that, we asked if the absence/lack of c-di-AMP also makes cells sensitive/resistant to any antibiotics, and we found that Δ disA mutant becomes sensitive to erythromycin, tobramycin, and moderately resistant to ciprofloxacin by diverse mechanisms. Altogether, our study highlights three different c-di-AMP-regulated pathways in *M. smegmatis* leading to differential drug sensitivity of the strain.

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Materials and methods

Bacterial strains, media, and growth conditions

Mycobacterium smegmatis mc²155 (WT strain), the knockout strains of Δ disA and Δ pde, and different overexpression strains of the bacteria were cultured in Middlebrook 7H9 broth (MB7H9, Hi-Media, India) at 37°C, containing 2% glucose, providing a carbon source, and 0.05% tween-80 (vol/vol). Agar (1.5% w/v) (HiMedia) was used to make MB7H9 agar plates. Antibiotics hygromycin and kanamycin were obtained from Sigma–Aldrich, USA, and added to the culture media at a final concentration of 50 and 25 µg/ml, respectively. Based on the assays, ciprofloxacin, erythromycin, and tobramycin were used at different concentrations, and the antibiotics were purchased from Sisco Research Laboratories (SRL), India, and Sigma–Aldrich, USA.

For the growth inhibition assay, bacteria were grown in MB7H9 till the mid-exponential phase, washed twice, and inoculated into M9 minimal media containing 1 mM MgCl₂, 0.3 mM CaCl₂ with 0.2% glucose (Gadagkar and Gopinathan 1980) by adjusting the final optical density at 600 nm (OD₆₀₀) of 0.03. Either erythromycin or tobramycin was added to the tubes containing 5 ml of bacterial cultures reaching the final concentrations of 10 and 0.2 µg/ml, respectively, and kept at 37°C shaking for 72 hours. Growth of the strains was estimated by measuring the optical density of the cultures (OD₆₀₀) at different time points and the final graph was made using GraphPad Prism. In the case of low-temperature growth assay, bacterial cultures were kept at 20°C shaking for 120 hours. Multiple biological replicates of each strain were used in all the growth assays.

Minimum inhibition concentration (resazurin microtiter assay) assay

Minimum inhibition concentration (MIC) values of the antibiotics were estimated using Resazurin microtiter assay (REMA) adapted from an earlier protocol (Palomino et al. 2002). Briefly, the OD₆₀₀ of the overnight-grown bacterial cultures was measured and the turbidity was adjusted to a McFarland standard 1, further, the cultures were diluted in 1:10 in fresh MB7H9 media containing glucose and tween. After that, 196 µl of the diluted culture of each strain was added to a single row of a 96-well microtiter plate (in duplicates) carrying a 2-fold (for regular MIC) or sometimes 1.15-fold (if we needed to find minor differences in MIC). The serial dilution of antibiotics at a specific range was chosen based on the strains used. After the culture was added to the wells, MIC plates were kept (standing) in a 37°C incubator for 36 hours. After that, 30 µl of freshly prepared resazurin dye (0.01%) was added to each well, and the plates were incubated for an additional 5–6 hours. The resazurin dye turned pink if bacterial growth was present in the wells, indicating a change from its original blue color. The minimum antibiotic concentration at which the resazurin dye remained blue was determined as the MIC.

Minimum bactericidal concentration determination

The minimum bactericidal concentration (MBC) of ciprofloxacin was determined against different strains as follows: bacterial strains were grown till the mid-exponential phase and the final OD₆₀₀ was adjusted to 0.2 by adding fresh MB7H9 media. 4 ml of culture was taken in different tubes with variable concentrations of ciprofloxacin at 2, 1, 1.5, and 0.5 µg/ml (final concentration) in each tube. Tubes were kept on a shaker incubator at 37°C and plating was done after 24 hours. Plates were then incubated for 3 to 4

days at 37°C to allow for the appearance of distinct colonies before being counted. The minimum drug concentration that was able to kill 99.9% of the bacterial population (3 logarithmic decreases) within 24 hours was recorded as the MBC of ciprofloxacin against that particular strain.

In vitro combination studies in broth

The combination MIC involving two antibiotics was performed using the checkerboard method (Balasubramanian et al. 2014). In a 96-well microtiter plate, 2 µl of the first drug was added vertically and 2 µl of the second drug was added horizontally (from the antibiotic strength of 100 times) to obtain multiple combinations of these two drugs in each well. 196 µl of OD₆₀₀-adjusted cultures (similar to MIC) were added to each well and the combination MIC plates were kept (standing) in a 37°C incubator for 36 hours. Next, 30 µl of freshly prepared resazurin dye (0.01%) was added to each well, and the plates were incubated for an additional 5–6 hours. To assess if drug combinations have an additive or synergistic effect, the fractional inhibitory concentration index (ΣFIC) was calculated using the formula: (MIC_{drug 1} in combination/MIC_{drug 1} alone) + (MIC_{drug 2} in combination/MIC_{drug 2} alone).

Disk diffusion assay

As previously described (Rawat et al. 2004), cultures were grown until the mid-exponential phase and adjusted the final OD₆₀₀ to 0.7 using fresh MB7H9 medium. From each culture, 100 µl was spread on MB agar plates and allowed to dry completely. Then, ciprofloxacin-impregnated (2.5 mg/ml) disks were kept at the center of the culture spread plates and the plates were kept at a 37°C incubator (without inverting). After 3 days, zone of inhibition (cm²) was calculated to estimate the antibiotic sensitivity.

Time-kill kinetics assay

As described earlier (Bag et al. 2023), the secondary cultures of *M. smegmatis* were grown till the mid-exponential phase with an ~OD₆₀₀ of 0.7. This was further diluted to 0.2 OD, which corresponds to ~2 × 10⁷ colony-forming units (CFU)/ml. 4 ml of each culture was taken to a glass tube along with 2.5 µg/ml (final concentration) of ciprofloxacin (10X MIC), and the tubes were incubated at 37°C with shaking. Plating was done at time 0 and at an interval of 2 hours for 8 hours or directly after 24 hours. Colonies were counted after 72 hours of incubation at 37°C for CFU/ml estimation. Three biological replicates of each strain were used for the assay.

EtBr efflux assay

The efflux assay protocol was adapted from the previously described method (Li et al. 2004, Pal and Ghosh 2022). Briefly, *M. smegmatis* secondary cultures were grown till OD₆₀₀ of ~0.8, after washing twice using PBS, the cells were resuspended in PBS and kept at 37°C shaker incubator for 1 hour to induce starvation. Next, CCCP (carbonyl cyanide m-chlorophenylhydrazone) with a final concentration of 100 µM was added to each tube and the tubes were incubated in a 37°C shaker incubator for 30 min. After that, ethidium bromide with a final concentration of 0.5 µg/ml was added and tubes were again incubated for 30 min. Cells were taken into wells of 96-well black plates after removing extracellular EtBr and glucose was then added to stimulate efflux activity. Subsequently, the EtBr fluorescence measurements were taken after 1 hour in the Varioskan Flash multimode reader (Thermo Fisher Scientific) [excitation 530 nm, emission 590 nm].

Quantitative real-time polymerase chain reaction (qRT-PCR)

As previously described (Singla et al. 2023), secondary cultures of *M. smegmatis* were grown till the mid-exponential phase ($OD_{600\text{ nm}}$ 0.7–0.8) in the MB7H9 medium containing 2% glucose, and 0.05% tween. Cultures were washed once using 1X PBS, followed by re-suspension in Trizol (Sigma), and further the cells were lysed by the bead-beating method. RNA isolation was performed using Qiagen's RNeasy Minikit. After the final step of RNA elution, the residual DNA was removed by DNase treatment followed by EDTA treatment and heat inactivation at 65°C. For the qRT-PCR assay, a one-step TB Green RT-PCR kit (Takara) was utilized and the assay was conducted in a Bio-Rad CFX Opus 96 Real-Time PCR machine. The Bio-Rad CFX Maestro software was used for gene expression analysis across samples and the mRNA levels of the gene of interest were normalized to the *sigA* house-keeping gene of *M. smegmatis*. The experiments were carried out in three biological replicates with two technical duplicates each time.

Expression and purification of Arr

For Arr overexpression, *Escherichia coli* BL21 Rosetta (DE3) strain was transformed with pET28-*arr* plasmid and cells were grown at 37°C until the $OD_{600} \sim 0.7$. Arr protein expression was induced by 1 mM IPTG for 16 hours in a 26°C shaking incubator. Further, the cell pellet was collected by centrifugation at 6000 rpm for 15 min. Sonication was used to lyse the cells resuspended in the lysis buffer containing 20 mM Tris and 300 mM NaCl with a pH of 6.5. The lysate was centrifuged at 13000 rpm for 90 min to remove cell debris. For Arr protein purification, the supernatant collected was added onto a 5 ml Ni²⁺-NTA Sepharose column (GE Healthcare) kept at 4°C with a flow rate of 0.5 ml/min. After this binding step, the wash buffer (20 mM Tris, 300 mM NaCl, 20 mM imidazole, pH 6.5) was passed through the column at a 2 ml/min flow rate to detach all the non-specifically bound proteins. The bound Arr protein was eluted using elution buffer containing 20 mM Tris, 300 mM NaCl, and 250 mM imidazole with a pH of 6.5, and passed at a 0.5 ml/min flow rate. The protein of interest in the elution fractions was visualized using SDS-PAGE. Fractions that contained the protein were pooled, concentrated, and subjected to size exclusion chromatography (SEC) purification using a Superdex 75 column (GE Healthcare) using SEC buffer (20 mM Tris, 150 mM NaCl, pH 6.5). For overexpression of uniformly ¹⁵N labeled Arr, cells were grown in minimal media (M9) with 1 g/l of ¹⁵NH₄Cl till $\sim 0.8 OD_{600\text{ nm}}$, followed by the addition of 0.5 mM Isopropyl β -D-1-thiogalactopyranoside (IPTG) and incubating it further for 16 hours at 16°C. The final protein sample for nuclear magnetic resonance (NMR) acquisition was prepared in a buffer containing 20 mM NaH₂PO₄ and 200 mM NaCl with pH 6.5.

Circular dichroism spectroscopy

Circular dichroism (CD) spectroscopy was used to evaluate the secondary structure of free Arr in solution. The spectra were recorded on a JASCO J-715 spectropolarimeter placed at 25°C, using a 1 cm path cuvette (Hellma Analytics) with a scanning speed of 100 nm/min and a response time of 4 sec. The scan was performed from wavelength 190 to 260 nm. A sample concentration of 15 μ M was used in the experiments. The average of each sample was calculated as the mean of three iterations, and a baseline correction was applied to eliminate buffer interference.

NMR spectroscopy

2D NMR spectrum (¹H-¹⁵N HSQC) of Arr protein was recorded on a Bruker 700 MHz instrument equipped with a room temperature probe at 298 K. The Arr sample was concentrated to 200 μ M and 10% D₂O was added for the spectrometer deuterium lock. For the titration experiments, increasing amounts of unlabeled compounds (NAD⁺ and cyclic-di-AMP) were added to a ¹⁵N labeled Arr solution, and the 2D spectra of (¹H-¹⁵N HSQC) were recorded after each step of titration. Data were processed using Bruker TOPSPIN 3.1 and analyzed using NMRFAM-SPARKY (Lee et al. 2015).

Microscale thermophoresis

Experiments to determine binding affinity were conducted using a Monolith NT.115 series instrument (Nano Temper Technologies GmbH). As per the manufacturer's protocol, the Arr protein was labeled using the Monolith Protein labeling kit RED-NHS second-generation amine (Nano Temper Technologies GmbH). Approximately 10 μ l of the sample was diluted with microscale thermophoresis (MST) buffer [containing 20 mM HEPES (pH 6.5) and 200 mM NaCl] and were loaded into regular capillaries of Monolith NT.115, and thermophoresis was measured for 30 sec. The Monolith software was used for analysis. To quantify K_d, the change in normalized fluorescence (F_{norm}, fluorescence after thermophoresis divided by initial fluorescence) was analyzed as a function of binding concentration. Curves for K_d data were fitted to a 4-parameter logistic equation using nonlinear regression in Palmist 1.5.6 software (Scheuermann et al. 2016) and plotted in Gussi software v1.1.0 (Brautigam 2015).

Arr activity assay

As described earlier (Zheng and Lupoli 2021), the *in vitro* assay of Arr was conducted by co-incubating the purified Arr enzyme (2 μ M) with the substrate rifampicin (15 μ M), cofactor NAD⁺ (25 μ M), and variable concentration of c-di-AMP (in a range of 0 to 45 μ M). The reaction mixtures were prepared in 100 μ l of 50 mM Tris-HCl buffer (pH 7.5) and incubated at 37°C for 30 min. The reaction was terminated by adding 100 μ l of 0.1 N formic acid solution, and the resulting mixture was analyzed using LCMS. The samples were analyzed using an ESI QTOF mass spectrometer (Bruker Daltonics, Germany) connected to the Agilent HPLC 1260. A reverse-phase C8 column with a plate number of 1 was utilized. The mobile phase consisted of water and acetonitrile with 0.1% formic acid, with a flow rate of 0.2 ml/min. A linear gradient ranging from 10% to 90% acetonitrile was used. The data was analyzed using Data Analysis version 4.1 software.

Polysome profiling

The method for the polysome profiling was adapted from a previous study (Agarwal et al. 2018). Briefly, *M. smegmatis* culture was grown in M9 media at 37°C or 20°C till $\sim 0.6 OD_{600\text{ nm}}$ and chilled on the ice-salt mix. The cultures were washed by centrifuging at 7000 rpm, 4°C for 5 min and the cells were resuspended in polysome buffer of pH-8 containing 20 mM Tris-HCl 10 mM MgCl₂, 60 mM NH₄Cl, 3 mM DTT, and 5% sucrose. The cell extract was prepared by beads beating and the cell lysates were taken for low-temperature (4°C) centrifugation at 13000 rpm for 40 min. Approximately 10–15 $OD_{260\text{ nm}}$ of cell extract was loaded on a linear sucrose gradient (10%–35%) prepared using BioComp gradient master. Samples were centrifuged using a Beckman SW41 rotor at 36000 rpm, 4°C for 3.5 hours. Gradients were fractionated using a BioComp gradient fractionator with continuous A₂₆₀ measurement.

In vitro GFP protein synthesis assay

The 70S ribosomes were purified from *M. smegmatis* WT, Δ disA, and Δ disA + pDisA strains following the laboratory protocol (Li et al. 2018). *Mycobacterium smegmatis* initiation factors IF1, IF2, and IF3 and elongation factors EF-Tu and EF-G were cloned in pET24a plasmid with (His)₆-tag in the C-terminus, overexpressed in *E. coli* BL21-CodonPlus (DE3)-RIPL strain, and purified to homogeneity using Ni-NTA affinity column connected to the AKTA-prime plus chromatography system (Degiacomi et al. 2016). For the *in vitro* protein synthesis assay, a reconstituted transcription-translation system was constructed that is composed of highly pure and active translation components from *M. smegmatis*, including 70S ribosomes, translation factors, bulk tRNA, and aminoacyl tRNA synthetases (Degiacomi et al. 2016). The reaction also contained amino acids, ATP, GTP, T7 RNA polymerase, and an optimized energy-regenerating system containing phosphoenol pyruvate (PEP), pyruvate kinase, and myokinase in HEPES-polymix buffer, pH 7.5 (Brautigam 2015). GFP synthesis was initiated by the addition of pET24-GFP plasmid to the reaction mix at 37°C. Synthesis of active GFP protein was followed by monitoring GFP fluorescence (512 nm) with time in a TECAN Infinite 200 PRO plate reader (Li et al. 2018, Parajuli et al. 2021).

Dipeptide formation assay

For the MF dipeptide formation assay, the initiation complex and an elongation mix were prepared. The initiation complex was formed by mixing 2 μ M 70S ribosome, 10 μ M MFF mRNA (transcribed *in vitro* with T7 RNA polymerase and purified with Sephadex G-75 gel filtration chromatography), 1 μ M [³H]-fMet-tRNA^{fMet}, charged and purified (Parajuli et al. 2021), and 2 μ M of all three initiation factors in HEPES-polymix buffer of pH 7.5. An elongation mix was prepared by incubating 40 μ M tRNA^{Phe}, 0.5 mM Phe, 1 unit ^{Phe}tRNA synthetase, and 40 μ M EF-Tu in the same buffer. The initiation complex and elongation mix were incubated separately at 37°C for 15 min. Dipeptide formation was initiated by mixing the two and the reaction was terminated after 10 sec by adding 17% formic acid. The [³H]-fMet and [³H]-fMet-Phe peptides were separated by reverse phase HPLC equipped in line with radioactivity detection (β -RAM model 3 IN/US systems) (Wang et al. 2020).

Results

c-di-AMP-mediated rifampicin sensitivity in *M. smegmatis* is Arr dependent

As reported earlier, *M. smegmatis* Δ pde strain showed higher sensitivity to rifampicin (inhibits bacterial RNA polymerase) compared to *M. smegmatis* WT by 8 fold (MIC: 2 μ g/ml), and in Δ pde + pPde strain, the MIC value reverts to WT level (16 μ g/ml) (Chaudhary et al. 2023). Conversely, we also observed that the overexpression of DisA (pMV261-disA) protein (c-di-AMP synthetase) results in an increase in rifampicin sensitivity (4 fold) in *M. smegmatis* confirming the potential role of high c-di-AMP concentration behind the phenotype (Table 1). DisA is a dual-functional protein, where 2 different domains of DisA take part in different activities: the DAC domain containing N-terminal synthesizes c-di-AMP and the HhH domain containing the C-terminal is involved in DNA scanning and repair in several bacteria (Witte et al. 2008, Zhang and He 2013). Hence, we wanted to confirm that it is c-di-AMP hyperaccumulation and not the DisA enzyme's overexpression itself that is responsible for rifampicin sensitivity. To prove that, we overexpressed the catalytically inactive mutant of DisA (D84A)

Table 1. List of MIC values (μ g/ml) of different *M. smegmatis* strains against rifampicin.

Strains	MIC (μ g/ml)
	Rifampicin
<i>Mycobacterium smegmatis</i> WT	16
<i>Mycobacterium smegmatis</i> Δ pde	2
<i>Mycobacterium smegmatis</i> Δ pde + pPde	16
<i>Mycobacterium smegmatis</i> Δ disA + pMV261 empty	16
<i>Mycobacterium smegmatis</i> Δ disA + pMV261-disA	4
<i>Mycobacterium smegmatis</i> Δ disA + pMV261-disA (D84A)	16
<i>Mycobacterium smegmatis</i> WT + pMV261-arr	32
<i>Mycobacterium smegmatis</i> Δ pde + pMV261 empty	1
<i>Mycobacterium smegmatis</i> Δ pde + pMV261-arr	32
<i>Mycobacterium smegmatis</i> Δ pde + pMV261-arr (D83A)	1
<i>Mycobacterium smegmatis</i> Δ arr + pMV261 empty	0.25
<i>Mycobacterium smegmatis</i> Δ arr + pMV261-disA	0.25
<i>Mycobacterium smegmatis</i> Δ arr + pMV261-disA (D84A)	0.25

in *M. smegmatis* Δ disA background and found no change in MIC compared to empty vector control (16 μ g/ml) (Table 1). As Arr is known to be involved in rifampicin ribosylation and acquiring partial resistance against rifampicin (Dabbs et al. 1995, Baysarowich et al. 2008, Zheng and Lupoli 2021), we hypothesized that c-di-AMP (at high concentration) could interfere with Arr (Msm_{Arr}) enzyme's function and thus make cells sensitive to rifampicin. To explore that, first, we overexpressed Arr protein (pMV261-arr) in Δ pde strain we found a 16-fold increase in the MIC value (from 2 to 32 μ g/ml). Next, we asked whether it was Arr's expression or catalytic activity that was inhibited by c-di-AMP; and the fact that RNA-seq data (Chaudhary et al. 2023) did not point out any significant difference in *arr* expression at the mRNA level in Δ pde strain (Log₂foldChange is -0.082), we focused on the Arr enzyme's catalytic activity and no such modulation in MIC value was observed with overexpression of catalytically inactive mutant (D83A) of *arr*. As expected, there was hardly any change in rifampicin MIC value with the *arr* overexpression (pMV261-arr) in the WT background (Table 1). To prove that the Arr is the key downstream effector molecule, we tested the rifampicin sensitivity of the *M. smegmatis* Δ arr + pMV261-disA strain, and we found, unlike the WT background, no difference in MIC was observed compared to blank vector control (Table 1). Both these observations directly imply the involvement of c-di-AMP with Arr, thus mediating the rifampicin sensitivity of the cells.

c-di-AMP interacts with Arr protein of *M. smegmatis* and prevents ribosylation of rifampicin

A structural comparison of c-di-AMP and NAD⁺ (Arr enzyme's substrate) revealed partial similarity (both the molecules share an adenine and two ribose sugar moieties in their structure), and we explored the possible binding of the c-di-AMP molecule to Arr enzyme's catalytic domain and thus inhibiting the ribosylation activity. The SEC profile of the purified Arr protein showed that the protein elutes as a monomer with a molecular weight of ~16 kDa (Fig. S1A). The eluted fractions were analyzed for purity by SDS-PAGE (Fig. S1B). Purified protein was further characterized using CD and NMR spectroscopy. CD spectrum of Arr showed that it is mainly α -helical in solution (Fig. S1C). This was confirmed by

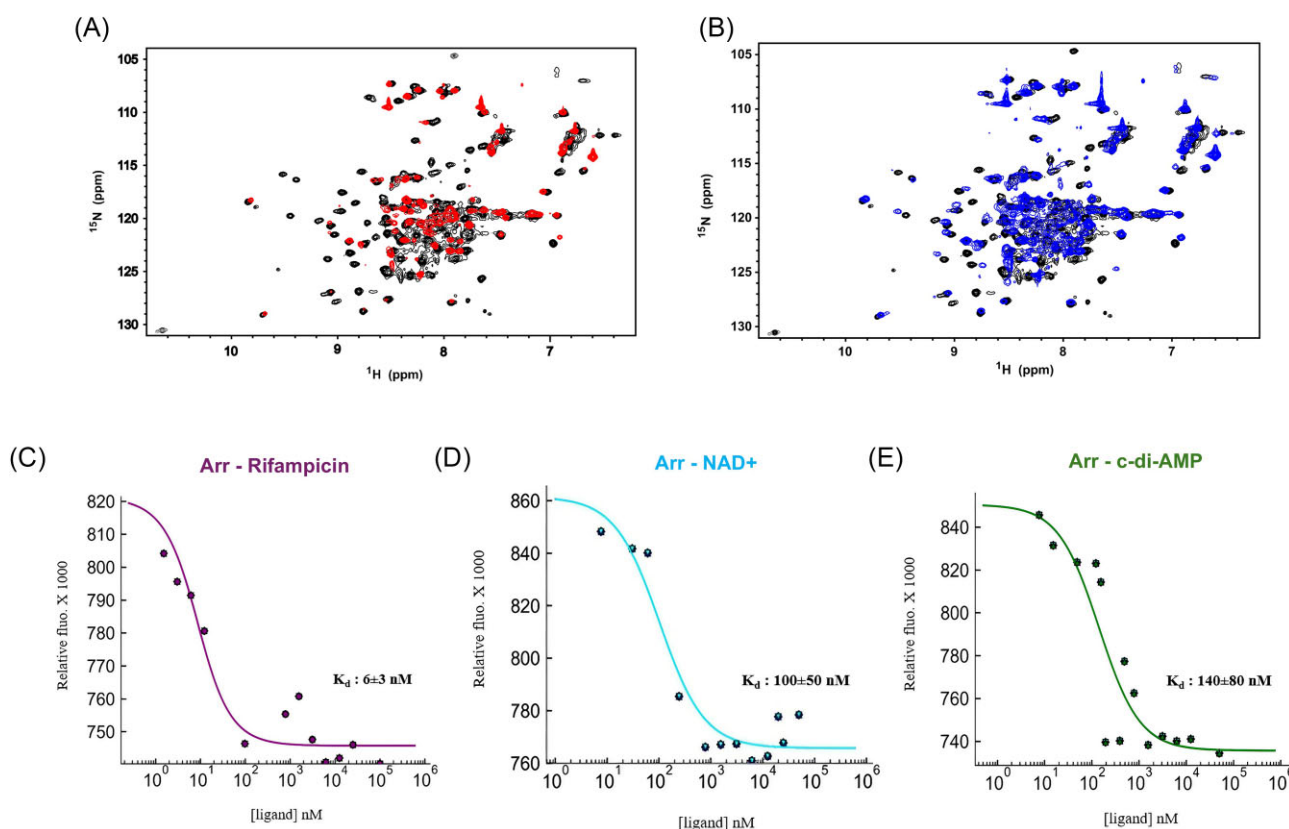


Figure 1. Biophysical enumeration of the interaction of Arr with c-di-AMP. NMR spectroscopic validation of interaction of Arr protein (A) with NAD⁺, where the overlay of 2D ¹H-¹⁵N HSQC spectra of free Arr (black) and Arr with NAD⁺ (red) at 1:2 molar ratio is shown and (B) with c-di-AMP; an overlay of 2D ¹H-¹⁵N HSQC spectra of free Arr (black) and Arr with c-di-AMP (blue) at 1:2 molar ratio is shown. Physical interaction of Arr with (C) rifampicin, (D) NAD⁺, and (E) c-di-AMP was confirmed by microscale thermophoresis (MST). Binding affinities in terms of dissociation constants (K_d) are also mentioned in individual cases.

recording a 2D ¹H-¹⁵N HSQC NMR spectrum of uniformly ¹⁵N labeled Arr, which showed a spectrum with well-dispersed resonance cross peaks indicating that the protein is well-ordered in solution (Fig. S1D). The interaction of Arr with NAD⁺ and c-di-AMP was monitored using NMR spectroscopy. For the NMR titrations, the unlabeled compounds were titrated into a solution of uniformly ¹⁵N labeled Arr, and a 2D ¹H-¹⁵N HSQC NMR spectrum was recorded at each step of titration (Fig. 1A and B). In the case of both NAD⁺ and c-di-AMP titrations, at a 1:2 protein-to-compound ratio, we observed resonance peak disappearances and/or chemical shift perturbations of several peaks in the 2D ¹H-¹⁵N HSQC NMR spectrum of Arr (Fig. 1A and B). This indicates that Arr interacts with both ligands. To further confirm and quantitate the interactions of Arr with Rifampicin (positive control), c-di-AMP, and NAD⁺, we used MST. The MST titrations showed that Arr interacts with Rifampicin, NAD⁺, and c-di-AMP with the apparent dissociation constants (K_d) of ~6 nM, 100 nM, and 140 nM, respectively (Fig. 1C, D, and E). Together, NMR and MST titration experiments unambiguously showed that c-di-AMP interacts with Arr with a comparable affinity of NAD⁺ (co-factor), which may result in the inhibition of the enzyme and preventing ribosylation of rifampicin. We proceeded to test the inhibition of Arr in the presence of c-di-AMP through LCMS analysis and estimated the formation of the modified rifampicin (ADP-ribosyl-rifampicin). Our findings indicated that the formation of ADP-ribosyl-rifampicin was significantly inhibited in the presence of c-di-AMP (in a dose-dependent manner), while the MS peak of the reaction product i.e. ADP-ribosyl-rifampicin was distinctly visible with a peak iden-

tity at m/z of ~1364 (Fig. 2A), which had been reported before (Baysarowich et al. 2008). Parallely, we could see that due to the inhibition of Arr enzyme by c-di-AMP, the substrate rifampicin with a peak identity at m/z of ~823 is unimpaired with increasing concentration of c-di-AMP (Fig. 2B). Next, based on the MS peak intensities of the product, we estimated the potency of the c-di-AMP inhibitor and found the approximate K_i (inhibitor constant) of the reaction to be ~24 μ M (Fig. S2).

LfrR protein expression is mediated by c-di-AMP resulting in ciprofloxacin sensitivity

We found that the Δ disA strain was moderately resistant to second-generation fluoroquinolone ciprofloxacin, which was proved in multiple assays. Previously, disk inhibition assay data clearly showed that Δ disA strain had a small yet significant resistance phenotype against ciprofloxacin and with the help of mutants, it was proved that the phenomenon is linked to c-di-AMP only, and not due to the absence of DisA enzyme's DNA scanning property (Chaudhary et al. 2023). We checked the increased resistance level of Δ disA strain by minimum inhibitory concentration (MIC) assay, where the difference of ciprofloxacin concentrations from one well to the next well is 1.15 fold instead of the standard 2 fold so that we could precisely measure the MIC value. Our data showed a modest increase (~1.85 fold) of MIC in Δ disA strain (0.869 μ g/ml) compared to WT (0.468 μ g/ml) (Table 2). Following this, we did an MBC (minimum bactericidal concentration) estimation of the strains and found the MBC value

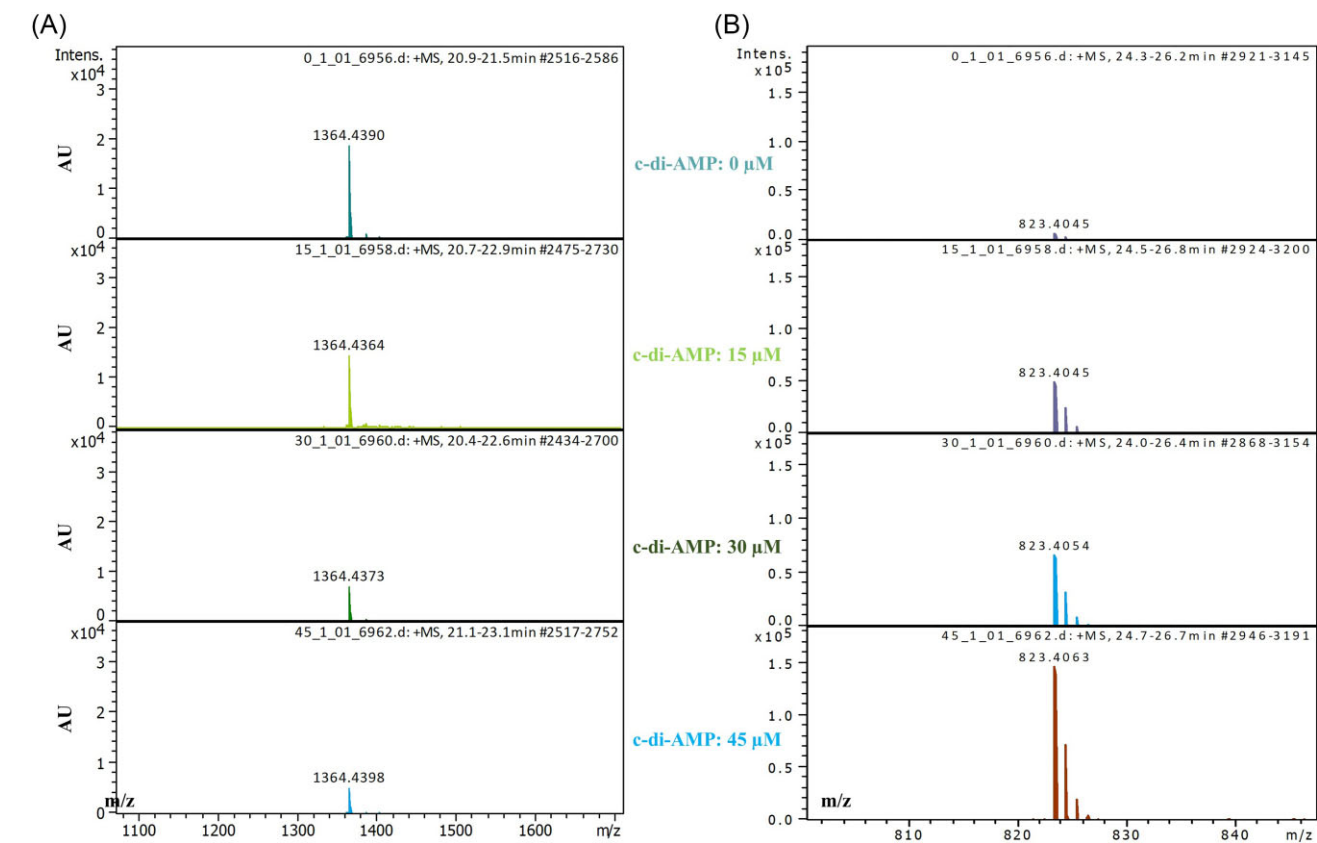


Figure 2. LCMS analysis of Arr reaction mixtures with variable concentrations of c-di-AMP. (A) The peak corresponding to the ADP-ribosylated rifampicin ($m/z \sim 1364$) was detected with gradually decreasing intensity as the concentration of c-di-AMP increased [0 μM , 15 μM , 30 μM , and 45 μM]. (B) As a confirmation of the Arr inhibition, the peak of the substrate rifampicin ($m/z \sim 823$) can be observed more intact with higher concentrations of c-di-AMP. [45 μM > 30 μM > 15 μM > 0 μM].

Table 2. List of MIC and MBC values ($\mu\text{g/ml}$) of different *M. smegmatis* strains against ciprofloxacin and erythromycin.

Strains	Ciprofloxacin
	MIC ($\mu\text{g/ml}$)
<i>Mycobacterium smegmatis</i> WT	0.468
<i>Mycobacterium smegmatis</i> ΔdisA	0.869
<i>Mycobacterium smegmatis</i> ΔdisA + pDisA	0.468
<i>Mycobacterium smegmatis</i> WT + pMV261 empty	0.468
<i>Mycobacterium smegmatis</i> ΔdisA + pMV261 empty	0.869
<i>Mycobacterium smegmatis</i> WT + pMV261- <i>lfrR</i>	0.395
<i>Mycobacterium smegmatis</i> ΔdisA + pMV261- <i>lfrR</i>	0.395
	MBC ($\mu\text{g/ml}$)
<i>Mycobacterium smegmatis</i> WT	1
<i>Mycobacterium smegmatis</i> ΔdisA	1.75
<i>Mycobacterium smegmatis</i> ΔdisA + pDisA	1
	Erythromycin
	MIC ($\mu\text{g/ml}$)
<i>Mycobacterium smegmatis</i> WT	48.4
<i>Mycobacterium smegmatis</i> ΔdisA	27.7
<i>Mycobacterium smegmatis</i> ΔdisA + pDisA	48.4

of ΔdisA (1.75 $\mu\text{g/ml}$) strain was higher than WT (1 $\mu\text{g/ml}$) strain (Table 2). Similarly, when we compared the survival percentage of cells after treatment with 2.5 $\mu\text{g/ml}$ ciprofloxacin (10X MIC) for 24 and 8 hours, the killing profile of ΔdisA strain showed a higher survival and better ciprofloxacin tolerance, respectively, compared to *M. smegmatis* WT (Fig. 3A and B). In all the cases, the complemented strain (ΔdisA + pDisA) showed a reversal of resistance to the WT level and further confirmed the direct link of ciprofloxacin tolerability with the presence or absence of c-di-AMP (Table 2). Next, the EtBr efflux assay (Fig. S3) suggested that ΔdisA strain had increased efflux pump activity, which correlates well with the moderately low level of resistance phenotype, unlike a target gene mutation (Pal and Ghosh 2022), where large shifts in MIC values are typically observed. Based on this, we shortlisted some candidate genes from the literature and one candidate that drew our attention was the *LfrR*-*LfrA* efflux module (Liu et al. 1996, Buroni et al. 2006). Once identified, we were specifically interested to know if the repressor gene *lfrR* was significantly downregulated in the absence of c-di-AMP. Indeed, the RNA-seq data (Chaudhary et al. 2023) revealed a significant fold change of expression (Log₂fold change: -1.378) of the *lfrR* repressor gene (MSMEG_6223, a TetR/AcrR family transcriptional regulator) in ΔdisA strain; later, the qRT-PCR data also confirmed significant downregulation of *lfrR* gene (Fig. 3C). Next, we overexpressed the *lfrR* gene from the pMV261 vector under a moderately

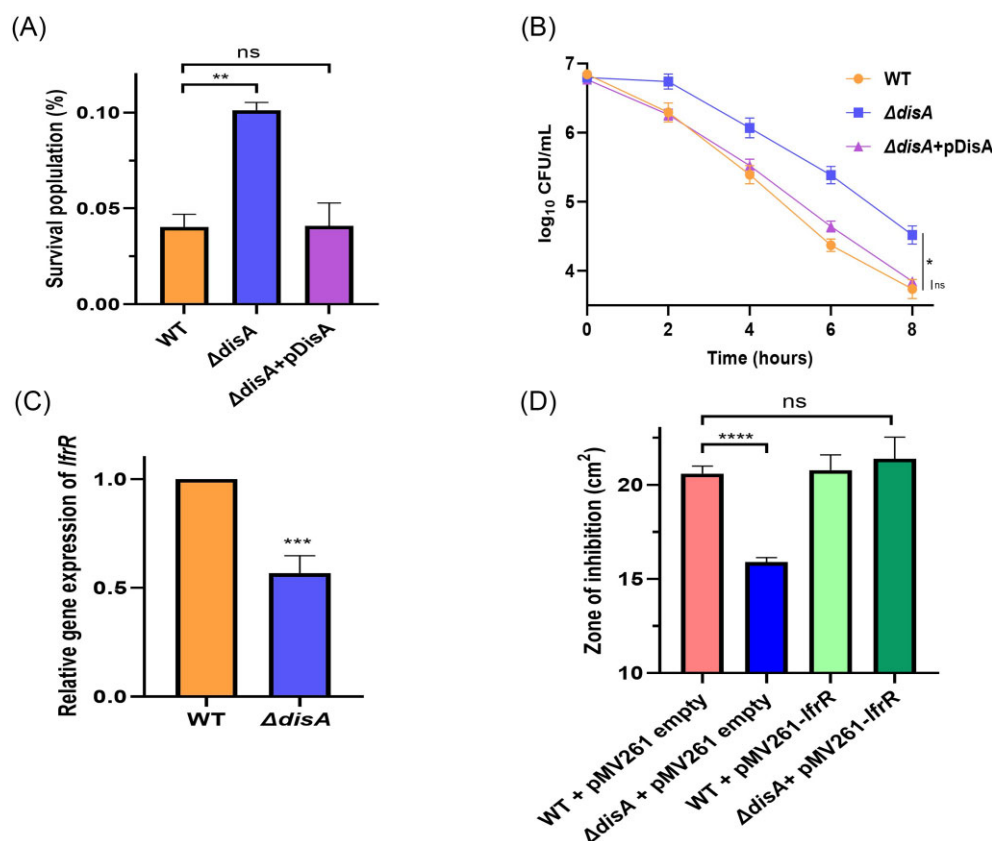


Figure 3. Elucidation of the role of c-di-AMP in ciprofloxacin resistance phenotype. The c-di-AMP null mutant strain *M. smegmatis* $\Delta disA$ shows more resistance to ciprofloxacin compared to *M. smegmatis* WT strain shown in multiple assays. (A) The survival percentage of *M. smegmatis* $\Delta disA$ strain was 2 fold higher than *M. smegmatis* WT strain after 2.5 μ g/ml ciprofloxacin (10X MIC) treatment for 24 hours, whereas the complemented strain $\Delta disA$ + pDisA showed a similar level of sensitivity to WT. (B) Time kill kinetics of *M. smegmatis* $\Delta disA$ strain shows an increased tolerance to ciprofloxacin compared to WT and complemented strain; an unpaired t-test was used at the final t-8 hours timepoint [T = 2]. (C) qRT-PCR validation of significant downregulation of *lfrR* gene (repressor) in *M. smegmatis* $\Delta disA$ strain, confirming the basis of low-level ciprofloxacin resistance. (D) Ciprofloxacin disk diffusion assay shows that the overexpression of *lfrR* repressor from multicopy pMV261 vector reverses the ciprofloxacin resistance phenotype in *M. smegmatis* $\Delta disA$ strain. All the graphs are plotted using GraphPad Prism 8 and an unpaired t-test was used to calculate statistical significance. *** = $P < 0.001$; ** = $P < 0.01$; * = $P < 0.05$; ns = non-significant.

strong constitutive *hsp60* promoter (P_{hsp60}) and, more importantly, without the native 5'-UTR, which might be responsible for c-di-AMP-mediated regulation, in $\Delta disA$ strain and found that the MIC value of *M. smegmatis* $\Delta disA$ + pLfrR strain dropped by ~2 fold to the WT level (0.395 μ g/ml). In identical conditions, the control strain *M. smegmatis* $\Delta disA$ + pEmpty retained the higher MIC (0.869 μ g/ml) (Table 2). Parallely, ciprofloxacin (2.5 μ g/ml) disk diffusion assay showed a significant increase in drug sensitivity for *M. smegmatis* $\Delta disA$ + pLfrR strain compared to *M. smegmatis* $\Delta disA$ + pEmpty strain (Fig. 3D). This particular result reinforces the importance of LfrR's role in repressing the *lfrRA* operon and increasing susceptibility to ciprofloxacin. Additionally, we examined the promoter expression of the *lfrRA* operon with a transcriptional reporter construct by fusing the upstream DNA sequence of *lfrR* to a promoterless-GFP vector pMN406- Δ Pimyc (Singla et al. 2023). The fluorescence readout of GFP in strains harboring the pMN406 Δ Pimyc-*lfr_US* reporter plasmid indeed showed an increase in promoter activity (Fig S4) in the absence of c-di-AMP ($\Delta disA$ strain compared to WT) elucidating the increased efflux of ciprofloxacin linked to downregulation of LfrR repressor. As a validation of the pMN406 Δ Pimyc-*lfrR* reporter construct, *M. smegmatis* pMN406 Δ Pimyc-*lfr_US* + pMV261-*lfrR* strain (overexpression of *lfrR* from a multi-copy plasmid) was used as a positive control where the promoter was found to be completely turned off and

no GFP fluorescence was detected (Fig S4). Altogether, our results suggested a potential role of c-di-AMP in regulating the LfrRA efflux activity of ciprofloxacin in *M. smegmatis*.

Ribosomal protein L7/L12 is downregulated in the absence of c-di-AMP leading to increased susceptibility to different translation inhibitors

We observed notable growth inhibition in the early and mid-exponential phase of growth of $\Delta disA$ strain in minimal media (M9) with protein synthesis inhibitors erythromycin (Fig. 4A) and tobramycin (Fig. 4B) at 10 μ g/ml and 0.2 μ g/ml, respectively. Next, we performed a MIC assay with erythromycin and found a ~1.75 fold difference in MIC value between WT and $\Delta disA$ strain, further confirming modest sensitivity phenotype against macrolide antibiotic erythromycin targeting 50S ribosomal subunit (Table 2). The complemented strain ($\Delta disA$ + pDisA) showed a reversal of sensitivity to the WT level and thus validated the bona fide connection of c-di-AMP with differential drug sensitivity. The third assay that confirmed increased susceptibility of $\Delta disA$ strain was Combi-MIC when we measured the MIC of one drug in the presence of another drug by using the checkerboard method (Balasubramanian et al. 2014) for the following combinations: Erythromycin + Chloramphenicol (combination I)

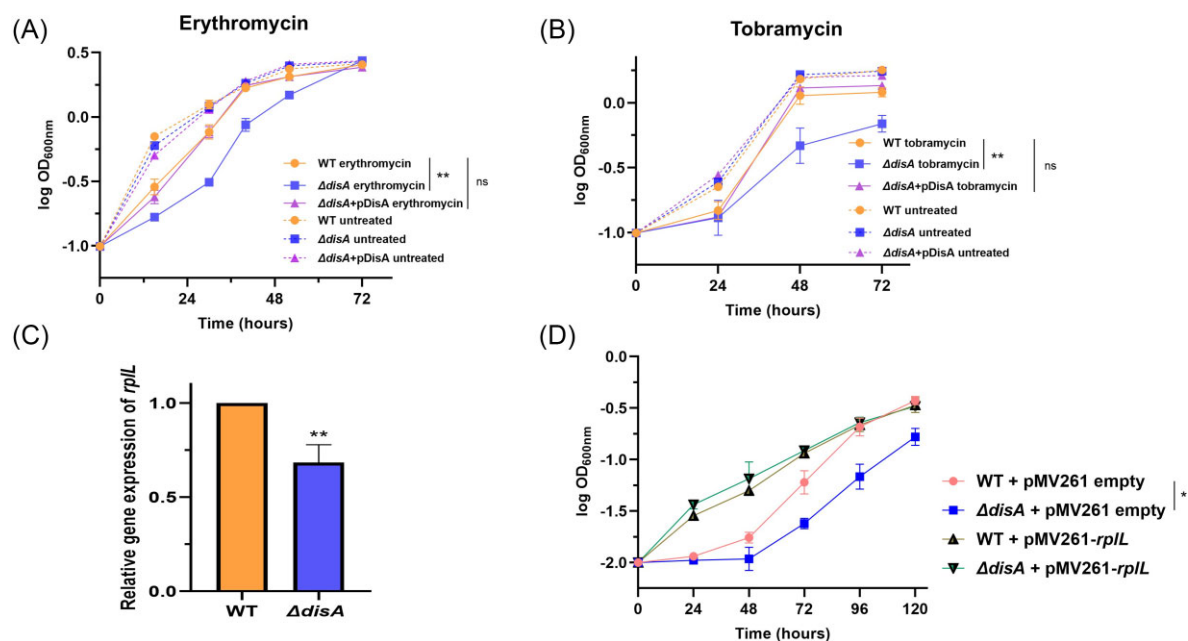


Figure 4. Lack of c-di-AMP makes Δ disA strain susceptible to multiple stresses. *Mycobacterium smegmatis* Δ disA strain showed increased susceptibility against two different protein synthesis inhibitors (A) erythromycin (macrolide) and (B) tobramycin (aminoglycoside) pointing toward a possible defect in ribosome structure and/or activity in the absence of c-di-AMP. The unpaired t-test was used at t-52 hours and t-72 hours time points for Figure 3A and B, respectively [T = 3]. (C) The qRT-PCR validation of significant downregulation of *rplL* gene (encodes for ribosomal protein L7/L12) in *M. smegmatis* Δ disA strain, reconfirming the observation from RNA-Seq analysis. (D) *Mycobacterium smegmatis* Δ disA strain was shown to have a possible ribosome-related growth defect at low temperature (20°C), which was recovered by overexpression of RplL protein from multicopy pMV261 vector. An unpaired t-test was used at the final t-120 hours time point [T = 2]. All the graphs are plotted using GraphPad Prism 8 and an unpaired t-test was used to calculate statistical significance. *** = $P < 0.001$; ** = $P < 0.01$; * = $P < 0.05$; ns = non-significant.

and Erythromycin + Streptomycin (combination II). To evaluate whether a particular combination resulted in an additive or synergistic outcome, the fractional inhibitory concentration index (Σ FIC) was calculated and found synergistic effects (as FIC indexes were < 0.5) of the drug combinations in both strains. However, the difference in Σ FIC values (Table S1) indicated the higher susceptibility phenotype of Δ disA strain against two combined protein synthesis inhibitors. In a recent study (Chaudhary et al. 2023), the differential expression of several ribosomal genes in Δ disA strain was highlighted by RNA-seq-based transcriptome analysis, and one of the most significant downregulated ribosomal genes was found to be *rplL* (codes for large subunit protein L7/L12). Subsequently, we performed qRT-PCR and confirmed significant downregulation (~40%) of *rplL* gene in Δ disA strain (Fig. 4C), which was earlier detected by RNA-seq analysis. To understand the biological effect possibly related to a defect in ribosomal biogenesis and/or assembly of subunits, we performed a growth curve analysis in minimal media at a low temperature (20°C) and it showed a clear growth defect in the Δ disA strain. Subsequently, we overexpressed *rplL* gene from multicopy pMV261 vector with constitutive hsp60 promoter and found that the low-temperature growth deficiency in Δ disA strain is recovered, suggesting that the loss of L7/L12 protein was responsible for the growth defect in the Δ disA strain (Fig. 4D). Similarly, it was also observed that *rplL* overexpression in Δ disA strain could alleviate the tobramycin sensitivity (Fig. S5). To comprehend the molecular mechanism behind the reduced expression of the L7/L12 protein in the Δ disA strain, we first conducted polysome profiling to investigate potential biogenesis irregularities and subunit imbalances in the ribosomes of the Δ disA strain. We found no difference between WT and Δ disA ribosomal subunits' profile when strains were grown at two different temperatures, suggesting Δ disA strain did not have significant ri-

bosome biogenesis defects (Fig. S6A and B). However, it could be possible that the observed difference could be attributed to ribosomal activity.

Ribosome extracted from Δ disA strain shows compromised activity in protein synthesis

The L7/L12 protein is important for efficient and accurate translation (Huang et al. 2010, Mandava et al. 2012, Ge et al. 2018). Since Δ disA ribosomes lead to an under-expression of the L7/L12 protein, a defect in translation could be anticipated. To identify the physiological inadequacy of the Δ disA ribosomes, we tested those in a full-length protein (GFP+) synthesis assay (Fig. 5A) using an *M. smegmatis*-based fully reconstituted transcription-translation system (Helgstrand et al. 2007, Degiacomi et al. 2016, Chen et al. 2020). As a control, we used ribosomes from the genetic equivalent WT and *disA*-complemented (Δ disA + pDisA) strains. Our results show that Δ disA ribosome is active in protein synthesis, but its activity is highly compromised as it could produce only 60% GFP + protein compared to the WT in a given time of 1 hour (Fig. 5B). The ribosomes from (Δ disA + pDisA) strain largely rescued the defect as it increased GFP + production to ~85% (Fig. 5B). To check further whether the defect in Δ disA ribosomes originates from the main catalytic step of protein synthesis, we have tested those in a dipeptide synthesis assay, which includes both decoding and peptide bond formation steps. The relative amount of the [3 H]-fMet mono-peptide and [3 H]-fMet-Phe dipeptides separated in reverse phase HPLC clearly shows less amount of (fMF) dipeptide formed with Δ disA ribosomes compared to the WT and Δ disA + pDisA ribosomes (Fig. 5C). Quantification of the dipeptide peaks suggests only ~54% dipeptide formed with Δ disA ribosomes (Fig. 5D), which matches well with its capacity in total protein syn-

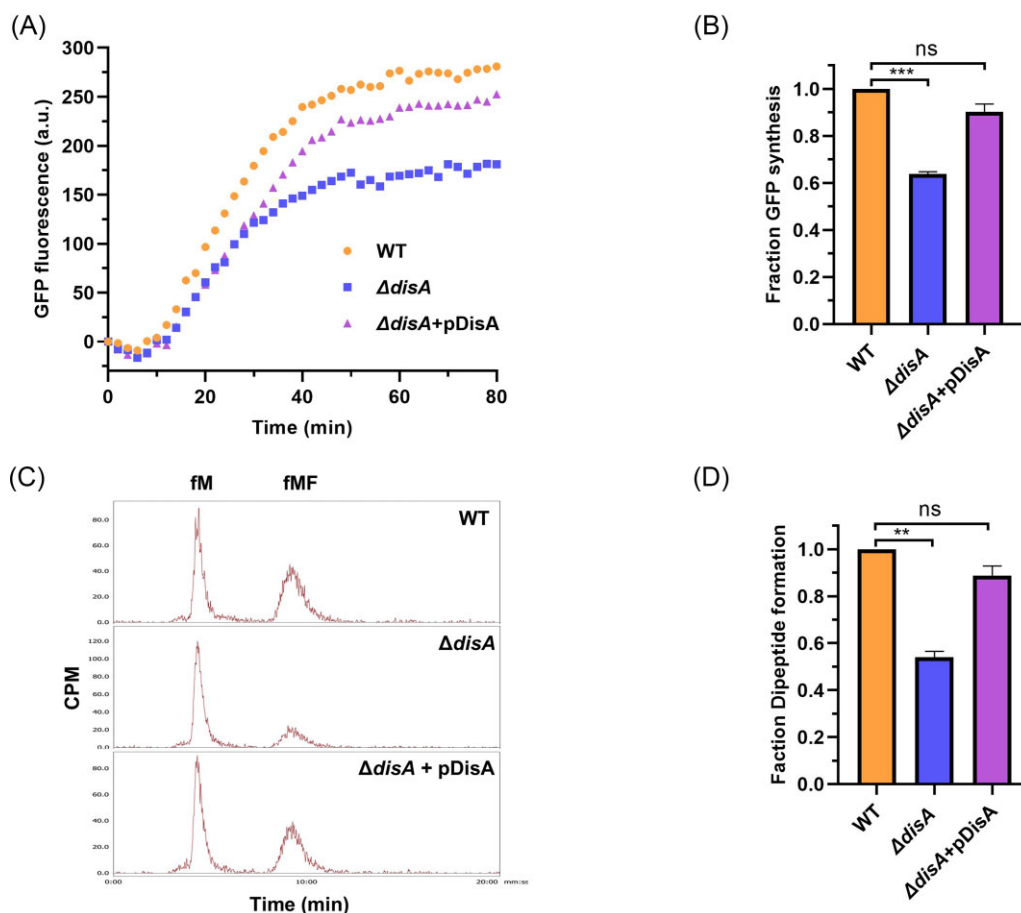


Figure 5. Compromised activity of the *ΔdisA* ribosomes in protein synthesis. (A) Time course of GFP + synthesis with isolated ribosomes from *M. smegmatis* WT, *ΔdisA*, and *ΔdisA* + pDisA strains in a fully reconstituted transcription-translation system, followed with fluorescence readout (512 nm) in real-time. (B) The fraction of GFP + synthesized in 1 hour with various ribosomes is labeled. (C) HPLC separation profiles of the mono (3H-fMet) and dipeptide (3H-fMet-Phe) with *ΔdisA*, WT, and *ΔdisA* + pDisA ribosomes. The relative fraction of the ([3H]-fM + [3H]-fMF) peak compared to [3H]-fMF peak indicated compromised elongation with *ΔdisA* ribosomes. (D) The fraction of dipeptide formed with the ribosome variants is indicated. The fraction of dipeptide formation was determined from the area percentage of fMF in total [3H] peak count ([3H]-fM + [3H]-fMF). All the graphs are plotted using GraphPad Prism 8, an unpaired t-test was used to calculate statistical significance. *** = $P < 0.001$; ** = $P < 0.01$; * = $P < 0.05$; ns = non-significant.

thesis. The ribosomes from *ΔdisA* + pDisA strain increased dipeptide production to ~87%, also in good agreement with its activity in GFP + synthesis (Fig. 5D). These results suggest that *ΔdisA* ribosomes are primarily defective in the elongation step of protein synthesis, which leads to the defect in total protein synthesis.

Discussion

Nucleotide messengers play a critical role in many bacteria, especially during stress adaptation phases. Previous studies have demonstrated a direct correlation between the intracellular ATP pool and the general effectiveness of antibiotics across various bacterial classes (Akhova and Tkachenko 2014, Lobritz et al. 2015, Huemer et al. 2020). Specifically in mycobacteria, a recent study proposed that the ATP bust is an important factor in antibiotic-induced cell death (Lodhiya et al. 2024). According to Manikandan et al. (2018) paper, it was found that c-di-AMP significantly contributes to the suppression of the SOS response in *M. smegmatis*. It achieves this by binding to the c-terminal motif of RecA protein and as well as causing translational repression of *recA* in the absence of c-di-AMP. The study also found that the *M. smegmatis* *ΔdisA* strain is susceptible to various DNA-damaging agents like MMS and MMC. However, the study does not provide any

sights into the antibiotic sensitivity in the absence of the RecA-mediated SOS response. Here, we show the mechanistic role of c-di-AMP in determining certain antibiotic tolerance phenotypes of *M. smegmatis*. Our results of c-di-AMP messenger controlling drug resistance phenotypes of a single organism against four different classes of antibiotics is a unique observation. In addition to that, we have done an in-depth investigation to find out the underlying mechanisms behind the differential drug sensitivity in a case-by-case manner (Fig. 6). The fact that c-di-AMP interacts with Msm_{Arr} enzyme and thus inhibits its rifampicin ribosylation activity, targeting c-di-AMP degradation enzyme Pde (by which the cells will have a higher intracellular c-di-AMP concentration) can be a logical approach to make the cells sensitive to rifampicin. Arr has been targeted before to augment rifampicin activity using specific libraries and an antibiotic-adjuvant combination has been shown to inhibit Arr making bacterial cells sensitive to rifampicin (Zheng and Lupoli 2021). Given this, our finding would be clinically significant to target ADP-ribosyltransferase MAB_0591 in pathogenic non-tuberculous bacteria like *M. abscessus* to make the organism more sensitive to rifampicin (Rominski et al. 2017).

One important finding of our study is the intrinsic modulation of efflux pump activity linked to intracellular c-di-AMP concentration in *M. smegmatis*, which could be a survival strategy under

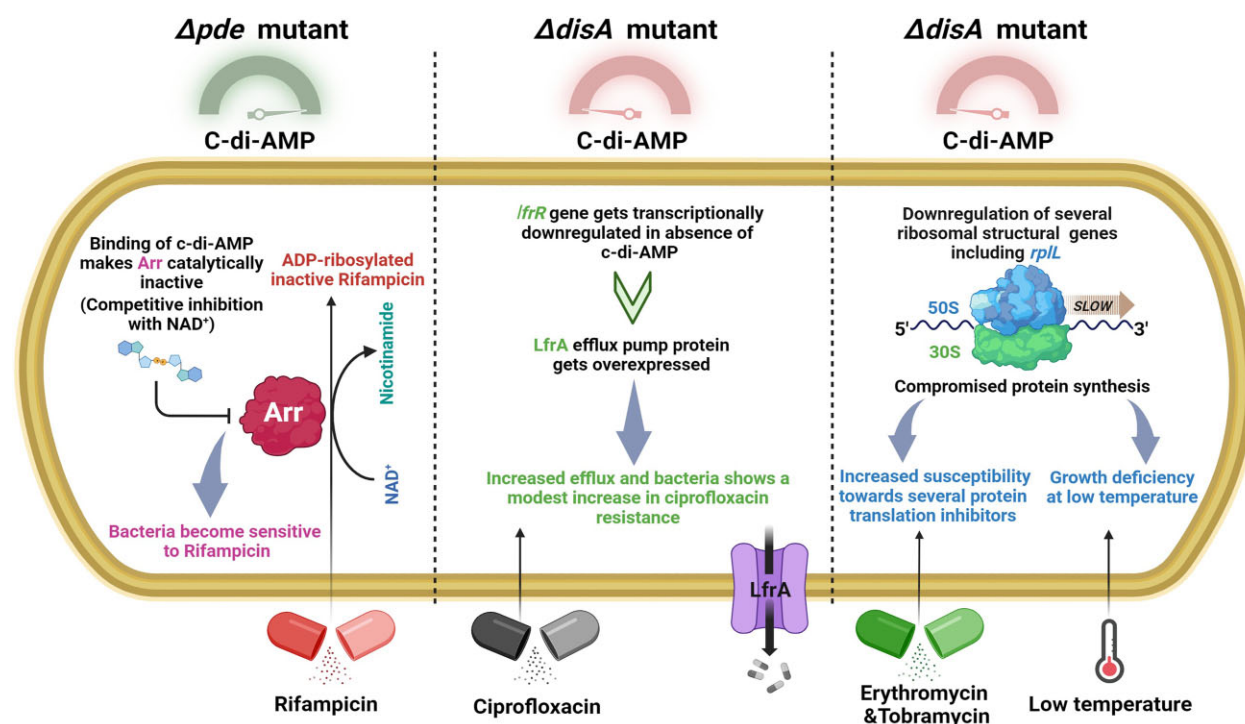


Figure 6. Schematic representation of our proposed model showing how c-di-AMP modulates multiple drug sensitivity pathways in *M. smegmatis* (created with BioRender.com).

transient fluoroquinolone exposure without accumulating any target mutation, especially when such mutation often comes with a huge fitness cost (Pal and Ghosh 2022). Another second messenger ppGpp was also shown recently to modulate efflux pump-related gene expression in *A. baumannii* (Jung et al. 2020) and result in MIC modulation of several antibiotics. We have shown how the presence and absence of c-di-AMP controls the expression of the efflux pump repressor gene *lfrR*. Further, we checked the presence of any secondary structures upstream of the *lfrR* gene by bioinformatics analysis and identified a putative stem-loop structure at the -35 region upstream of the *lfrR* gene (data not shown), which possibly blocks RNA polymerase's access to the promoter necessary to drive transcription of *LfrR*. Future studies should be aimed at understanding the mechanistic role of c-di-AMP based on possible disruption of stem-loop, resulting in optimal expression of *lfrR*.

Finally, we reported that the lack of c-di-AMP significantly downregulates the expression of the key ribosomal stalk protein L7/L12 in *M. smegmatis*, which leads to a compromised activity of the ribosomes in protein synthesis, hypersensitivity of the strain to low temperature, and to several antibiotics targeting protein synthesis (translation) machinery. The Δ *disA* ribosomes are defective in the elongation step of protein synthesis, which makes the total protein synthesis relatively slower. Possibly the slower protein synthesis widens up the time window for binding of the antibiotics to the Δ *disA* ribosomes, thereby making those more sensitive to the ribosome targeting antibiotics. A second messenger molecule modulating ribosome structure and function is a novel observation in any bacteria and would possibly open up a new field or research in the context of targeting such auxiliary pathways such as c-di-AMP synthesis for novel antibiotic research.

Altogether, our study has delineated the specific genes and pathways regulated by c-di-AMP at the transcriptional and post-

translational levels. Though c-di-AMP was discovered many years back (Witte et al. 2008), an elaborate study of this novel second messenger's important role in dictating antibiotic resistance and sensitivity phenotypes in a mechanism-driven manner is still lacking. The results presented in this paper related to c-di-AMP-driven differential antibiotic sensitivity response mechanisms in the model organism *M. smegmatis* gave novel insights that will provide further advancement in this field of second messenger signaling research. Since the c-di-AMP has been shown to have a definite link with *M. tuberculosis* virulence and triggering an immune response of the host (Yang et al. 2014, Ning et al. 2019), this study on the model organism *M. smegmatis* could give an ideal starting point to shape the future research related to the c-di-AMP and antibiotic resistance in *M. tuberculosis*. It is worth investigating whether c-di-AMP has a role in the drug sensitivity of *M. tuberculosis*, since its genome has corresponding orthologs for *disA* (Rv3586), *pde* (Rv2837c), and *rplL* (Rv0652) genes. On the other hand, the absence of orthologs for *arr* and *lfrR* in *M. tuberculosis* suggests that there may be alternative mechanisms for susceptibility to rifampicin and ciprofloxacin, respectively, in this bacterium. Further studies are required to elucidate these mechanisms and to develop new strategies to combat tuberculosis.

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Author contributions

A.P. and A.G. conceived the study. A.P., S.S., M.S., and A.G. designed the research; A.P., D.G., B.S., X.G., A.S., V.C. R.J., and A.G. performed experiments; A.P., S.S., U.V., M.S., and A.G. analyzed the data; A.P., S.S., M.S., and A.G. wrote the paper.

Supplementary data

Supplementary data is available at [FEMSLE Journal](https://femsle.journalonline.com) online.

Conflict of interest: The authors declare no conflict of interests.

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